

SURAPANENI HEMA SREE

Data Engineer

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Smyrna, GA

Professional Summary

- Data Engineer with 3+ years of experience building scalable ETL/ELT pipelines, cloud data platforms, and analytics-driven solutions across healthcare and banking domains.
- Hands-on in **PySpark, Python, SQL, SparkSQL, Airflow, Databricks, Snowflake, ADF, AWS Glue**, and big-data processing frameworks.
- Skilled in designing and optimizing **data models, Lakehouse architectures, Delta Lake, and analytical datasets** used for ML and BI workloads.
- Strong experience with **predictive modeling, feature engineering, and ML workflows** supporting healthcare risk analytics and financial fraud use cases.
- Proficient in building dashboards and analytical insights using **Power BI, Tableau, and big-data SQL engines**.
- Known for high-quality delivery, reliability, and strong collaboration with data scientists, engineering teams, clinicians, and business stakeholders.

Professional Experience

UnitedHealth Group

Jan 2025 – Present

Data Engineer

Atlanta, GA

- Develop and maintain **PySpark + Databricks** pipelines processing 10M+ daily clinical and claims records for care management and utilization insights.
- Built **Bronze–Silver–Gold Lakehouse layers** enabling actuarial models, predictive risk scoring, and population health analytics.
- Created automated ETL workflows using **Airflow, ADF, and Delta Lake**, improving refresh cadences and data reliability.
- Designed data models for **HEDIS quality metrics, readmission risk prediction, and claim anomaly detection** used by clinical leadership.
- Optimized Spark workloads through caching, cluster autoscaling, broadcast joins, and optimized storage formats; improved runtime by 40%.
- Collaborated with data scientists to integrate ML models into production pipelines using **MLflow + feature store**.

HDFC Bank – Enterprise Data Engineering

May 2022 – July 2024

Data Engineer

Hyderabad, India

- Developed **PySpark + AWS Glue** ETL processes for credit card, loan, and customer-behavior datasets supporting fraud and credit risk analytics.
- Implemented partitioned Parquet datasets on S3 enabling faster downstream reporting and ML query performance.
- Built automated quality checks using **Great Expectations** and SQL-based validation layers.
- Integrated upstream systems (CBS, LOS, payments) into centralized banking data lake.
- Collaborated with BI teams to deliver curated dashboards for delinquency patterns, fraud alerts, and portfolio performance.

Projects

AI-Driven Healthcare Readmission Prediction System

Python • PySpark • MLflow • Databricks • Healthcare Analytics

- Built a predictive ML model using clinical, claims, and encounter datasets to identify high-risk patients for 30-day readmission.
- Engineered 100+ features including comorbidity scores, utilization history, medication adherence, and encounter characteristics.
- Integrated model into production ETL pipeline using Databricks Jobs + MLflow for scheduled scoring and dataset refresh.

Financial Fraud Detection Engine – Real-Time Pipeline

Kafka • Spark Streaming • Python • AWS Glue • S3

- Designed a streaming pipeline detecting abnormal transactional behavior using statistical anomaly detection + ML models.
- Processed millions of events per day using Kafka + Spark Streaming with automated alerts for suspicious activities.
- Delivered dashboards enabling rapid fraud mitigation and actionable insights.

Technical Skills

- **Big Data:** PySpark, Spark SQL, Databricks, Delta Lake, EMR
- **Cloud:** AWS, Azure
- **ETL/ELT:** ADF, Airflow, Glue, Dataflow
- **Databases:** Snowflake, SQL Server, PostgreSQL, MySQL
- **Languages:** Python, SQL, Scala (basic)
- **ML/AI:** Feature engineering, MLflow, predictive modeling, anomaly detection
- **BI Tools:** Power BI, Tableau

Education

Kennesaw State University <i>Master's in Science in Computer Science</i>	Kennesaw, GA <i>Aug 2024 - Dec 2025</i>
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